Credit and funding adjustments

When determining the fair value of an instrument, it may be necessary to record adjustments to the Firm's estimates of fair value in order to reflect counterparty credit quality, the Firm's own creditworthiness, and the impact of funding:

- Credit valuation adjustments ("CVA") are taken to reflect the credit quality of a counterparty in the valuation of derivatives. CVA are necessary when the market price (or parameter) is not indicative of the credit quality of the counterparty. As few classes of derivative contracts are listed on an exchange, derivative positions are predominantly valued using models that use as their basis observable market parameters. An adjustment therefore may be necessary to reflect the credit quality of each derivative counterparty to arrive at fair value.

  The Firm estimates derivatives CVA using a scenario analysis to estimate the expected credit exposure across all of the Firm's positions with each counterparty, and then estimates losses as a result of a counterparty credit event. The key inputs to this methodology are (i) the expected positive exposure to each counterparty based on a simulation that assumes the current population of existing derivatives with each counterparty remains unchanged and considers contractual factors designed to mitigate the Firm's credit exposure, such as collateral and legal rights of offset; (ii) the probability of a default event occurring for each counterparty, as derived from observed or estimated CDS spreads; and (iii) estimated recovery rates implied by CDS, adjusted to consider the differences in recovery rates as a derivative creditor relative to those reflected in CDS spreads, which generally reflect senior unsecured creditor risk. As such, the Firm estimates derivatives CVA relative to the relevant benchmark interest rate.

- DVA is taken to reflect the credit quality of the Firm in the valuation of liabilities measured at fair value. The DVA calculation methodology is generally consistent with the CVA methodology described above and incorporates JPMorgan Chase's credit spread as observed through the CDS market to estimate the probability of default and loss given default as a result of a systemic event affecting the Firm. Structured notes DVA is estimated using the current fair value of the structured note as the exposure amount, and is otherwise consistent with the derivative DVA methodology.

- The Firm incorporates the impact of funding in its valuation estimates where there is evidence that a market participant in the principal market would incorporate it in a transfer of the instrument. As a result, the fair value of collateralized derivatives is estimated by discounting expected future cash flows at the relevant overnight indexed swap ("OIS") rate given the underlying collateral agreement with the counterparty. Effective in 2013, the Firm implemented a FVA framework to incorporate the impact of funding into its valuation estimates for uncollateralized (including partially collateralized) over-the-counter ("OTC") derivatives and structured notes. The Firm's FVA framework leverages its existing CVA and DVA calculation methodologies, and considers the fact that the Firm's own credit risk is a significant component of funding costs. The key inputs are: (i) the expected funding requirements arising from the Firm's positions with each counterparty and collateral arrangements; (ii) for assets, the estimated market funding cost in the principal market; and (iii) for liabilities, the hypothetical market funding cost for a transfer to a market participant with a similar credit standing as the Firm.

Upon the implementation of the FVA framework in 2013, the Firm recorded a one time $1.5 billion loss in principal transactions revenue that was recorded in the CIB. While the FVA framework applies to both assets and liabilities, the loss on implementation largely related to uncollateralized derivative receivables given that the impact of the Firm's own credit risk, which is a significant component of funding costs, was already incorporated in the valuation of liabilities through the application of DVA.

The following table provides the credit and funding adjustments, excluding the effect of any associated hedging activities, reflected within the Consolidated balance sheets as of the dates indicated.

December 31, (in millions)	**2014**	2013
Derivative receivables balance[(a)]	**$ 78,975**	$ 65,759
Derivative payables balance[(a)]	**71,116**	57,314
Derivatives CVA[(b)]	**(2,674)**	(2,352)
Derivatives DVA and FVA[(b)(c)]	**(380)**	(322)
Structured notes balance [(a)(d)]	**53,772**	48,808
Structured notes DVA and FVA[(b)(e)]	**1,152**	952

(a) Balances are presented net of applicable CVA and DVA/FVA.
(b) Positive CVA and DVA/FVA represent amounts that increased receivable balances or decreased payable balances; negative CVA and DVA/FVA represent amounts that decreased receivable balances or increased payable balances.
(c) At December 31, 2014 and 2013, included derivatives DVA of $714 million and $715 million, respectively.
(d) Structured notes are predominantly financial instruments containing embedded derivatives that are measured at fair value based on the Firm's election under the fair value option. At December 31, 2014 and 2013, included $943 million and $1.1 billion, respectively, of financial instruments with no embedded derivative for which the fair value option has also been elected. For further information on these elections, see Note 4.
(e) At December 31, 2014 and 2013, included structured notes DVA of $1.4 billion and $1.4 billion, respectively.